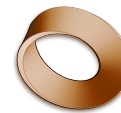


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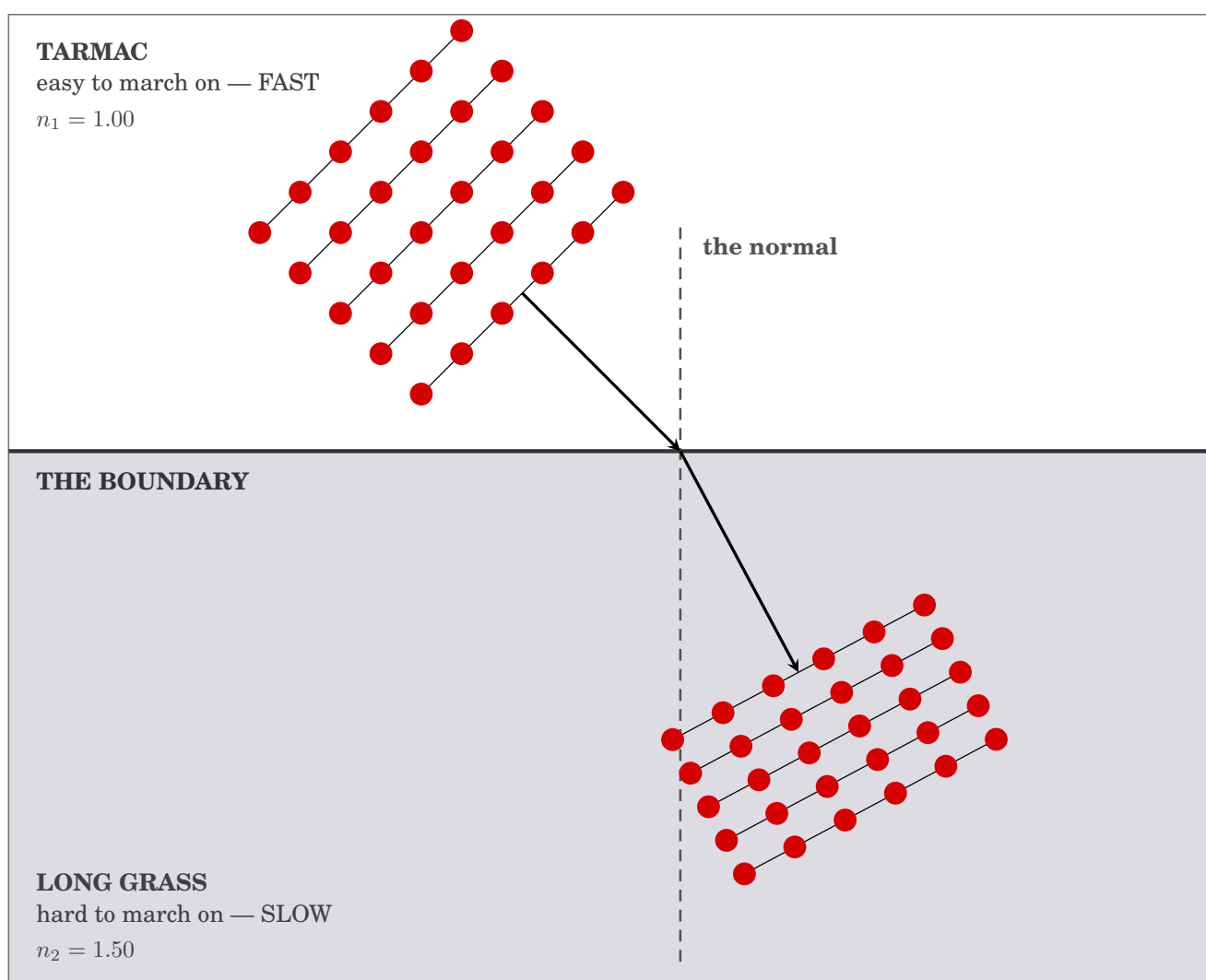
1.4 – Refraction I

Unit 1: Light & EM Spectrum

Learning intentions

Light changes direction when it crosses from one material into another. Today we find out **why** it does that — by becoming the light ourselves.

A marching band is crossing a field. They are on **tarmac**, where marching is easy and they move quickly. Ahead of them is **long grass**, where marching is hard and they can only move slowly. They reach the edge *at an angle*.



Outside

Line up in ranks, arms linked, and march at a steady pace towards the boundary *at an angle*. When your foot reaches the grass, take **smaller steps** — and **do not let go of the person next to you**. Watch what happens to the line.

1. The words we will use

These are the words you need to describe what you are about to see. You will use them again in every lesson on refraction, so keep this page.

Refraction

Light **changing direction** as it crosses from one material into another. It happens because the light changes **speed**.

Medium

The material the light is travelling through. Air, water and glass are all media. (One medium, two media.)

Boundary

The surface where one medium ends and the next one begins. This is the **only** place the light changes direction — never before it, never after it.

The normal

A line drawn at exactly 90° to the boundary, at the point where the ray hits it. Every angle in this topic is measured **from the normal**, never from the boundary itself.

Incident ray

The ray **arriving** at the boundary. “Incident” just means incoming.

Angle of incidence θ_1

The angle between the **incident ray** and the **normal**.

Refracted ray

The ray **after** it has crossed the boundary and changed direction.

Angle of refraction θ_2

The angle between the **refracted ray** and the **normal**.

Speed of light c

The speed of light in a vacuum: $c = 3.00 \times 10^8$ m/s — three hundred million metres every second. Nothing travels faster. Light is **slower than this in every material**.

Index of refraction n

A number telling you **how much slower** light travels in a material.

$$n = \frac{\text{speed of light in a vacuum}}{\text{speed of light in the medium}} = \frac{300\,000\,000 \text{ m/s}}{\text{speed of light in the medium}}$$

Air $n = 1.00$, water $n = 1.33$, glass $n = 1.50$. A **bigger n** means **slower light**.

2. Back inside.

(a) Turn back to **page 1**. In **pencil**, draw the band once it is in the long grass — at least three rows, each joined with a line as above.

(b) Now write down what happened. Which way did the band turn, and **why**? Use the words from page 2.

*Going into the slower medium the band turns **towards the normal**, because the side that reaches the grass first slows down first while the other side is still moving quickly — so the line swings round. The ranks also end up **closer together**, because everyone is taking smaller steps.*

3. Using n . Light travels through these three materials:

water $n = 1.33$ glass $n = 1.50$ diamond $n = 2.42$

(a) In which one does light travel **slowest**? How do you know?

Diamond. It has the biggest n , and a bigger n means slower light.

Extension

Rearrange the formula for n to give the speed of light *in* a material:

$$\text{speed in the medium} = \frac{300\,000\,000 \text{ m/s}}{n}$$

Now work out the speed of light in each one. Give your answers in m/s.

water ($n = 1.33$)	$\underline{2.26 \times 10^8} \text{ m/s}$
glass ($n = 1.50$)	$\underline{2.00 \times 10^8} \text{ m/s}$
diamond ($n = 2.42$)	$\underline{1.24 \times 10^8} \text{ m/s}$

(b) Coming in from air, in which one would the band turn the **most**?

Diamond again — it causes the biggest change in speed, so the ranks swing round the most.